**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 154 · Lesson 154 · Search Engines: Crawling, Indexing and Retrieval

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. Why does a server recheck a browser form?

Q2. When can binary search be used?

Q3. What data structure follows LIFO?

Q4. What is a database foreign key?

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

crawler — Search agent fetching linked pages.

inverted index — Terms mapped to document identifiers.

tokenisation — Breaking text into searchable units.

ranking — Ordering candidate search results.

## New learning

Discover — Crawler fetches pages found through links and sitemaps.

Index — Map each searchable term to candidate pages.

Retrieve — Match query terms and rank relevant results.

## Worked example

Page A has 'binary search' and Page B 'binary tree'. Build a small inverted index.

1. Tokenise both pages — Identify the input or diagram and predict the first result.

2. Map terms to page IDs — Follow the second stage and explain its importance.

3. Match search → A — binary→A,B; search→A; tree→B. The query 'binary search' intersects both postings and retrieves A.

# We Do · Practise with guidance

Page C has 'encryption key' and Page D 'encryption hash'. Which page answers 'encryption hash'?

Index terms

Identify the input or diagram and predict the first result.

Intersect term postings

Follow the second stage and explain its importance.

Return D

Inverted index: encryption→C,D and hash→D. Intersecting returns D. A search engine may rank by further signals after matching.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Explain crawling and indexing. [6]

### Question 2

Index P='cat dog', Q='dog bird' and identify result for 'dog bird'. [6]

### Question 3

Discuss why a new page might not appear in search and why robots.txt is not a security control. [8]

# Exit Ticket · Answer before checking

Search indexing and result ranking are the same operation.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Client checks are bypassable.

Q2. On sorted data.

Q3. Stack.

Q4. A reference to a key in another table.

### You Do pages · Independent questions

Q1. Crawler discovers/downloads pages [3], index stores searchable mapping [3].

Q2. cat→P, dog→P,Q, bird→Q [4]; Q intersection [2].

Q3. Discovery delays [2], indexing rules [2], update/crawl frequency [2], robots suggestions do not enforce access control [2].

### Exit Ticket · Model answer

False. Index identifies candidates; ranking orders them.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**